

Distinguishing AI-Generated and Real Images Using Swin-Transformer

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Abstract:

The recent advancements in generative AI technology have enabled the effortless creation of realistic synthetic images and videos, exacerbating issues related to privacy breaches and the generation of inappropriate content. Such synthetic content can lead to various societal challenges, including violations of privacy, defamation, and the dissemination of false information, necessitating urgent development of technical solutions for detection and prevention. This study proposes a novel detection model leveraging the shifted window mechanism and hierarchical structure of the Swin-Transformer to significantly enhance the capability to distinguish between AI-generated and real images. The proposed model was trained and validated using the RVF10k and Real and Fake Face Detection datasets to evaluate its generalization performance across diverse types of synthetic images. The model's detection capabilities were further strengthened by employing preprocessing and augmentation techniques such as Gaussian Blurring, Canny Edge Detection, Sobel Filter, and Laplacian Filter. Experimental results demonstrate that the proposed model outperforms existing CNN and Transformer-based models, reliably detecting even expertly manipulated synthetic images. This study establishes a robust technical foundation to address the misuse of generative AI, aiming to further enhance the model's generalization capabilities through future expansions to various synthetic media types and the development of lightweight models capable of real-time detection. By doing so, it seeks to provide effective countermeasures against privacy breaches and the creation of inappropriate content caused by generative AI, promoting its safe utilization and contributing to the restoration of societal trust.

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Introduction

The rapid advancements in generative AI, particularly in image generation, have introduced profound ethical and legal challenges. As technologies for producing highly realistic synthetic images continue to evolve, concerns surrounding their misuse, such as privacy violations and the creation of harmful or inappropriate content, have grown significantly. These challenges include the unauthorized use of personal appearances in explicit or defamatory media, leading to psychological harm, defamation, and legal disputes. Addressing these issues is a pressing task that impacts social safety and ethical responsibility [1].



REAL

FAKE



**FIGURE 1**

Example of Real and Ai-generated Image

The widespread integration of AI into various industries amplifies the potential harm of synthetic content. Deepfake technologies, in particular, pose severe risks such as the spread of misinformation, harassment, and the erosion of societal trust. The development of robust detection systems has thus become essential for ensuring the safe application of generative AI. Existing solutions, such as embedding watermarks in synthetic images or using adversarial training to enhance model robustness, provide foundational approaches but often fall short in addressing increasingly sophisticated manipulations [2]– [6].

To meet these challenges, this study proposes a novel detection model based on the Swin-Transformer [8]. Unlike traditional convolutional neural networks (CNNs) or early transformer-based models like Vision Transformer (ViT) [12], the Swin-Transformer leverages a hierarchical structure and a shifted window mechanism, making it particularly effective in detecting fine-grained manipulations within images. This model is tailored to distinguish between AI-generated and real images, focusing on subtle differences in patterns, textures, and artifacts that are often indicative of synthetic content.

The proposed model was trained and validated using the RVF10k [9] and Real and Fake Face Detection [10] datasets. The RVF10k dataset provides 10,000 high-resolution labeled images, enabling supervised learning, while the Real and Fake Face Detection dataset includes human-edited synthetic images, allowing the model to generalize to diverse types of manipulation. Various preprocessing techniques, such as Gaussian Blurring, Canny Edge Detection, Sobel Filters, and Laplacian Filters, were incorporated to enhance the model's ability to detect subtle features in synthetic images [13]– [16].

Experimental results demonstrate that the proposed Swin-Transformer-based model outperforms existing CNN and transformer-based detection models. The model achieved approximately 1.8% higher accuracy than its counterparts, highlighting the efficacy of its hierarchical structure and shifted window mechanism. This study contributes to addressing the societal risks posed by generative AI by providing a robust technical solution for distinguishing between real and AI-generated content.

The findings underscore the importance of developing advanced detection systems that can adapt to increasingly sophisticated synthetic media. By ensuring the safe use of generative AI, this research aims to restore societal trust and promote ethical applications of these technologies. Future work will focus on expanding the model's capabilities to cover other synthetic media types and developing lightweight, real-time detection models for broader accessibility.

Related Works

The rapid advancement of generative AI technologies has brought significant societal and ethical challenges, particularly with the rise of DeepFake technologies. DeepFakes, which produce highly realistic synthetic content such as manipulated faces or voices, initially gained traction in creative fields like filmmaking and entertainment. However, their misuse has led to grave consequences, including the unauthorized use of personal images in explicit or defamatory content, causing privacy violations, psychological harm, and legal disputes. Furthermore, DeepFake videos have been weaponized for spreading misinformation, political propaganda, and harassment, exacerbating the erosion of societal trust and sparking debates about the ethical implications of generative AI [1]– [6].

To combat these challenges, researchers have developed various methods to detect DeepFake and other AI-generated content. Early approaches relied on analyzing metadata or identifying pixel-level anomalies to distinguish between real and synthetic media. Generative adversarial network (GAN)-based detection models were also explored to identify generation patterns by training discriminators against synthetic outputs. While these methods initially showed promise, the continuous evolution of generative models has revealed their limitations, as newer synthetic content becomes increasingly indistinguishable from real images [7].

Convolutional Neural Networks (CNNs) have served as a robust baseline for image detection tasks. Notable architectures such as ResNet [11], DenseNet [17], InceptionV3 [18], and EfficientNet [19] have demonstrated remarkable capabilities in learning visual features. ResNet introduced residual connections to mitigate the vanishing gradient problem, enabling stable training in deeper networks. DenseNet further optimized efficiency by connecting each layer directly to all preceding layers, reducing redundant computations and improving gradient flow. InceptionV3's modular design allowed for effective multi-scale feature extraction, while EfficientNet employed neural architecture search to balance accuracy and

computational efficiency. Despite these advancements, CNN-based models often struggle to capture global dependencies and subtle differences in synthetic media, particularly as AI-generated content becomes more sophisticated.

Transformer-based models, originally developed for natural language processing, have reshaped the field of computer vision by introducing mechanisms for learning global dependencies within images. Vision Transformer (ViT) [12] demonstrated the potential of treating images as sequences of fixed-size patches, leveraging self-attention mechanisms to model the relationships between them. This global feature-learning approach allowed ViT to surpass traditional CNNs in tasks like image classification and object detection. However, ViT's reliance on large datasets and its limited ability to learn local features posed significant challenges, particularly for tasks requiring fine-grained analysis of synthetic image manipulations.

The Swin-Transformer [8] addressed these limitations by introducing a hierarchical structure and a shifted window mechanism, making it particularly effective for tasks such as synthetic image detection. The hierarchical structure processes images at multiple scales, enabling the model to analyze both fine-grained details and broader patterns. This multi-scale capability is especially advantageous for detecting subtle manipulations in generative images. Additionally, the shifted window mechanism bridges gaps in local information by facilitating interaction between adjacent regions, ensuring that local features are integrated into the global context.

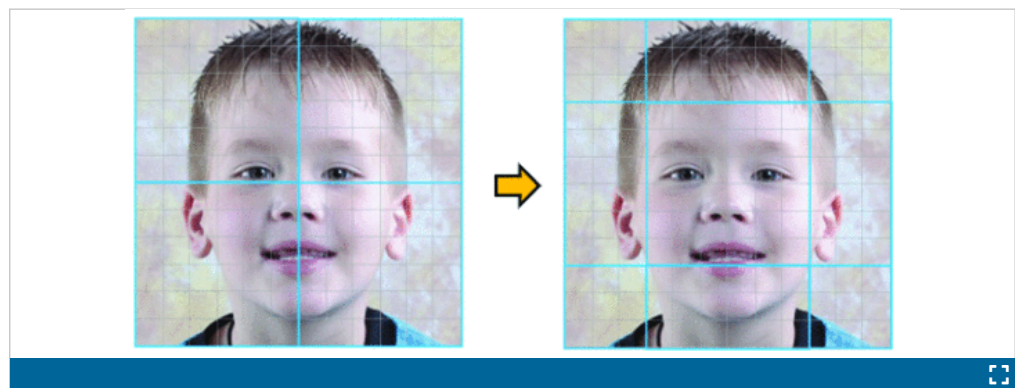


FIGURE II

Shifted Window Mechanism in the Swin-Transformer

The shifted window mechanism plays a crucial role in the Swin-Transformer's ability to capture both localized and global patterns. Unlike conventional models that process non-overlapping image segments, the shifted window mechanism introduces overlap between neighboring patches, enabling richer feature representation. This approach ensures that contextual information is shared between adjacent regions, making the model particularly effective at identifying subtle artifacts and inconsistencies in synthetic images. For example, in a facial image, this mechanism allows the model to link texture information from one region to structural features in another, such as combining skin texture details with geometric patterns in the eyes or mouth. This integration is critical for distinguishing real images from synthetic ones, where subtle inconsistencies may otherwise go unnoticed.

Preprocessing Techniques have also played a vital role in enhancing model performance. Techniques such as Gaussian Blurring [13], Canny Edge Detection [14], Sobel Filters [15], and Laplacian Filters [16] have proven effective in highlighting critical features and improving clarity. Gaussian Blurring reduces noise by smoothing high-frequency components, allowing the model to focus on primary structures. Canny Edge Detection extracts precise boundaries, aiding in the identification of artificial edges present in synthetic media. Sobel Filters detect edge gradients, emphasizing subtle boundary differences, while Laplacian Filters highlight abrupt intensity changes, effectively identifying irregular patterns in generative images. Together, these techniques enhance the ability of detection models to discern subtle manipulations.

This study builds upon these advancements by integrating preprocessing techniques with the Swin-Transformer architecture, leveraging its hierarchical and shifted window mechanisms. By combining the strengths of advanced preprocessing and the Swin-Transformer's multi-scale feature learning, the proposed model achieves state-of-the-art performance in distinguishing AI-generated images from real ones. These contributions address the pressing need for robust detection systems capable of adapting to the increasing sophistication of generative AI. This research not only advances the technical understanding of synthetic image detection but also provides a practical solution to mitigate the societal risks posed by generative AI misuse.

Our Methods

In this study, we propose a new model leveraging and modifying the strengths of the Swin-Transformer [8] to distinguish between real and generative images. While preserving the foundational principles of the Swin-Transformer [8], the model is restructured and optimized for the Real vs. Fake task to enhance analytical performance. The key design elements of the proposed model are as follows:

The proposed model modifies the window-based localized focus concept of the Swin-Transformer [8] to emphasize detailed analysis of specific parts of an image. Typically, processing the entire image may overlook subtle details such as minor artifacts in generative images. To address this, the input image is divided into multiple windows, with each window designed to intensively learn localized features. This approach highlights the details of each part, making it possible to clearly distinguish fine differences between generative and real images.

Additionally, localized focus is advantageous for detecting generative patterns that only appear in specific regions of the image. For instance, artificial features in generative images may appear in skin textures or subtle patterns in the background. The model conducts independent learning for each window to capture such subtle features. This ensures that localized characteristics contribute to the overall authenticity analysis while preserving computational resources without compromising performance.

The shifted window technique, a core concept of the Swin-Transformer [8], plays a crucial role in exchanging and integrating information between windows. In this study, the shifted window method is adapted to allow the information independently obtained from each window to be integrated with other windows, enabling richer feature representation. This approach helps the model effectively learn interrelated characteristics between generative and real images.

The shifted window technique facilitates the transfer of information from a specific region to adjacent windows, making it advantageous for establishing relationships between localized features. For example, in a facial image, if one window captures skin texture information, another window can use this information to analyze details of the eyes, nose, or mouth. Through this process, the model enhances contextual understanding of the entire image and effectively detects generative features present throughout.

Furthermore, the shifted window method integrates various parts of an image organically, maintaining the focused information of individual windows without loss. This is particularly critical for contrasting the attributes of real and fake images, allowing the precise integration of diverse information patterns through inter-window interactions.

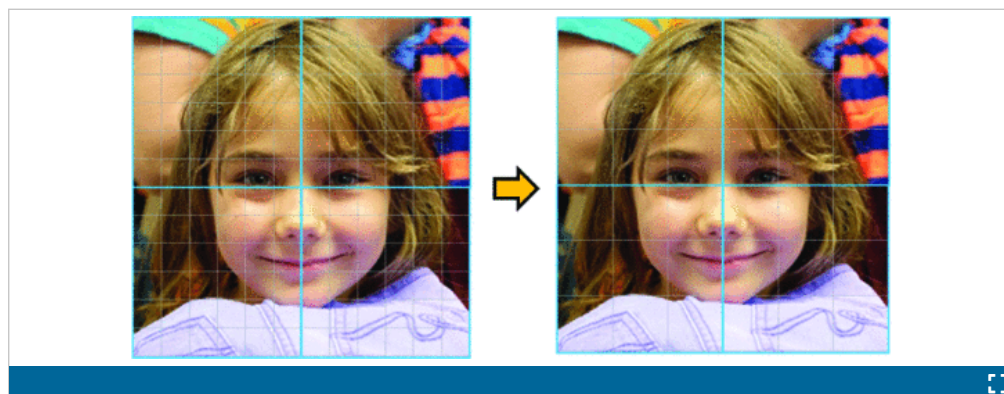


FIGURE III

Patch Merging Mechanism for Combine Local And Global Features.

The proposed model inherits the hierarchical structure of Swin-Transformer [8] while optimizing it for generative image analysis tasks. Swin-Transformer's hierarchical structure effectively extracts information across various scales, starting from small patches and progressively expanding to larger patches.

As illustrated in [FIGURE III], adjacent patches are merged into larger patches, enabling the model to calculate over a broader area of the image. This hierarchical approach allows the model to learn fine details at smaller scales and global composition at larger scales, making it well-suited for capturing features that differentiate real and generative images. For instance, small patches can focus on textures or skin details, while larger patches analyze symmetry or contours. This multi-scale approach is particularly effective for detecting abnormal patterns or irregularities often found in generative images.

The hierarchical structure also enhances learning efficiency and ensures consistent performance across images of varying resolutions. Considering that generative images can appear at various resolutions, this is a critical factor for the Real vs. Fake Task. The ability to analyze images regardless of resolution ensures robust performance under diverse conditions.

The proposed model incorporates flexible resolution handling to effectively process images of varying resolutions, enhancing the versatility of Swin-Transformer. This enables the analysis of real vs. generative images across resolutions ranging from high to low.

Since the quantity and quality of information vary by resolution, the ability to adapt is crucial. The model efficiently extracts unique features regardless of resolution, aiding in detecting subtle patterns differentiating generative and real images. For low-resolution images, the model emphasizes primary outlines and basic structures, while for high-resolution images, it focuses on textures and fine details. This adaptability ensures stable performance even in environments with mixed-resolution images.

Experiments

The experiments were conducted using a computational setup equipped with an Intel Core i9-13900H CPU, an NVIDIA GeForce RTX 4070 laptop GPU, and 32GB of RAM. The primary aim was to evaluate the performance of the proposed Swin-Transformer-based model in detecting AI-generated images and distinguishing them from real images. Two datasets, RVF10k [9] and the Real and Fake Face Detection dataset [10], were employed to test the model's robustness across diverse types of synthetic manipulations. The model's performance was compared against several state-of-the-art baselines, including ResNet-50 [11], DenseNet-121 [17], InceptionV3 [18], EfficientNet-Bo [19], and ViT-B/16 [12]. Metrics such as training loss, validation loss, accuracy, and resource utilization were used for evaluation.

I. Dataset and Preprocessing

The RVF10k dataset [9] consists of 10,000 labeled high-resolution images categorized as real or fake. This dataset is particularly valuable for training and evaluating detection models because it incorporates a wide range of facial features and synthetic manipulation styles. To ensure effective training, a comprehensive preprocessing pipeline was applied. Images were resized to a standard resolution of 256×256 and center-cropped to 224×224 to focus on the most relevant regions. Random horizontal and vertical flips were introduced to create variability and improve robustness against orientation-based biases. Additional augmentations, such as random color jittering and random erasing, were applied to simulate diverse lighting conditions and occlusions, enhancing the model's ability to generalize to unseen data [13]–[16].

The Real and Fake Face Detection dataset [10] complements RVF10k by including expertly manipulated images generated using advanced editing tools like Photoshop. These images represent a distinct challenge as they involve targeted modifications to specific facial features, such as eyes, mouth, or skin texture. Preprocessing for this dataset mirrored that of RVF10k, ensuring consistency across datasets while allowing the model to learn from diverse manipulations beyond those typically produced by GAN-based techniques.

II. Performance on Real vs Fake Face 10k Dataset

TABLE I Performace Comparision about RVF10k Dataset

Model	Train Loss	Train Accuracy (%)	Validation Loss	Validation Accuracy (%)
ResNet-50 [11]	0.12	85.3	0.21	88.4
DenseNet-121 [17]	0.10	88.7	0.19	90.2
InceptionV3 [18]	0.09	90.5	0.17	91.6
EfficientNet-B0 [19]	0.11	87.1	0.23	89.1
ViT-B/16 [12]	0.07	92.4	0.15	92.7
Ours	0.05	94.2	0.13	94.9

On the RVF10k dataset [9], the proposed Swin-Transformer-based model achieved the highest validation accuracy and the lowest validation loss among all models tested. ResNet-50 [11], one of the earlier CNN architectures, struggled to effectively capture the nuanced differences between real and synthetic images, resulting in lower accuracy and higher validation loss. DenseNet-121 [17], with its more efficient connectivity pattern, showed improved performance over ResNet-50, but it still fell short of models with attention

mechanisms. InceptionV3 [18], which employs multi-scale feature extraction through inception modules, demonstrated better results, particularly in scenarios with diverse manipulations.

ViT-B/16 [12] leveraged its self-attention mechanism to analyze global dependencies within image patches, achieving significant improvements over CNN-based models. However, the proposed Swin-Transformer-based model outperformed ViT-B/16 by incorporating hierarchical structures and shifted window mechanisms. These architectural features allowed the model to effectively integrate local details with global patterns, resulting in superior generalization and detection capabilities across diverse synthetic manipulations.

III. Performance on Real and Fake Face Detection Dataset

TABLE II Performace Comparision about Real and Fake Face Detection Dataset

Model	Test Loss	Test Accuracy (%)
ResNet-50 [11]	0.24	82.6
DenseNet-121 [17]	0.22	85.4
InceptionV3 [18]	0.20	86.9
EfficientNet-B0 [19]	0.26	80.3
ViT-B/16 [12]	0.18	88.7
Ours	0.16	91.2

The Real and Fake Face Detection dataset [10] posed unique challenges due to its inclusion of manually edited images, which often feature subtler manipulations than those produced by GANs. The proposed model demonstrated exceptional performance, achieving the highest test accuracy and the lowest test loss compared to all baselines. While ResNet-50 [11] and DenseNet-121 [17] struggled to generalize effectively to these types of manipulations, InceptionV3 [18] and EfficientNet-Bo [19] exhibited moderate success due to their ability to capture fine-grained features. ViT-B/16 [12] performed strongly, showcasing the benefits of self-attention in learning global relationships. However, the proposed model outperformed ViT-B/16, as its shifted window mechanism facilitated better integration of adjacent regions, enabling more accurate detection of subtle modifications.

IV. Performance on Real and Fake Face Detection Dataset

TABLE III Comparisional Resource Evaluation by Model

Model	Params (Miillion)	Memory (MB)	GFLOPs
ResNet-50 [11]	25.6	98	4.1
DenseNet-121 [17]	7.9	32	2.8
InceptionV3 [18]	23.9	92	5.7
EfficientNet-B0 [19]	5.3	20	0.4
ViT-B/16 [12]	86.6	335	17.6
Ours	28.3	110	1.2

Resource efficiency is a critical consideration for real-world applications, particularly in resource-constrained environments. The proposed Swin-Transformer-based model maintained a strong balance between performance and efficiency. With 28.3 million parameters and a computational cost of 1.2 GFLOPs, the model was more lightweight than ViT-B/16 [12] while significantly outperforming it in accuracy. Compared to EfficientNet-Bo [19], which is highly efficient but less accurate, the proposed model offered a superior trade-off between computational cost and performance. This balance of efficiency and performance highlights the practicality of the proposed model for deployment in mobile devices or edge computing systems where hardware resources are limited.

The experimental results underscore the effectiveness of the proposed Swin-Transformer-based model in detecting AI-generated images. On the RVF10k dataset [9], the model achieved significant improvements in accuracy and loss metrics, demonstrating its ability to handle diverse synthetic manipulations. Similarly, on the Real and Fake Face Detection dataset [10], the model excelled in distinguishing manually edited images, further validating its robustness. Its efficient architecture, combining hierarchical and shifted window mechanisms [8], enabled it to outperform all baseline models while maintaining competitive resource efficiency. These findings position the model as a reliable and practical solution for mitigating the risks associated with generative AI in real-world scenarios.

Conclusion

The increasing prevalence of AI-generated images, particularly in the form of DeepFakes, has underscored the need for robust detection systems capable of distinguishing synthetic content from real images. This study addresses these challenges by proposing a Swin-Transformer-based model optimized for the detection of AI-generated images. Leveraging the Swin-Transformer's hierarchical structure and shifted window mechanism [8], the model achieves superior performance by effectively capturing both local and global features, a critical capability for identifying subtle manipulations in synthetic media.

The experimental results validate the proposed model's effectiveness. On the RVF10k dataset [9], the model demonstrated state-of-the-art performance, achieving the highest validation accuracy and the lowest loss compared to baseline models, including ResNet-50 [11], DenseNet-121 [17], InceptionV3 [18], EfficientNet-Bo [19], and ViT-B/16 [12]. The model's ability to generalize to diverse manipulations was further confirmed on the Real and Fake Face Detection dataset [10], where it excelled in distinguishing real images from manually edited synthetic ones, a task often more complex than detecting GAN-generated content.

In addition to its accuracy, the proposed model offers competitive resource efficiency. With 28.3 million parameters and a computational cost of 1.2 GFLOPs, the model balances performance and practicality, making it suitable for deployment in resource-constrained environments. This efficiency extends its applicability to mobile devices, edge computing platforms, and real-time detection scenarios, broadening access to advanced detection tools in both developed and underserved regions.

The study also highlights the importance of preprocessing techniques [13]– [16], such as Gaussian Blurring, Canny Edge Detection, and Sobel Filters, in enhancing the model's ability to identify subtle differences between real and synthetic images. By emphasizing critical features, these techniques further improve the model's robustness and detection capabilities.

Future research will focus on expanding the model's scope to other forms of synthetic content, including audio and video DeepFakes, as well as exploring lightweight variants of the model for broader accessibility. Additionally, integrating multi-modal data, such as combining image-based analysis with metadata or contextual information, could further enhance detection accuracy and applicability across diverse scenarios.

In conclusion, the proposed Swin-Transformer-based model provides a robust and efficient solution to the pressing issue of detecting AI-generated images. By achieving state-of-the-art performance and maintaining resource efficiency, the model offers a scalable and practical tool for mitigating the societal and ethical risks posed by generative AI. This research represents a significant step toward ensuring the responsible and ethical use of AI technologies in an increasingly digital world.

Authors	▼
Figures	▼
References	▼
Citations	▼
Keywords	▼
Metrics	▼